Beyond the Blob: Localizing Inserted Lesions with Aligned Vessel Forecasts

Anonymous CVPR submission

Abstract

In chest CT, a blood vessel that bends or branches inside a candidate region can look like a small round lesion. A detector may then accept the region for the wrong reason, or point to the vessel instead of the lesion. We ask whether predicting the vessels from the surrounding tissue helps such a detector, and whether it helps because the prediction is in the right place*. We build pairs of CT crops that are identical outside a 14-mm ball, one with and one without an inserted lesion, so that a vessel forecast fitted to the outside is shared by both. As a control, we rotate the same forecast by 90 degrees, which keeps its values but breaks its alignment with the anatomy. At a matched 10% false-positive rate, the aligned forecast put the detector's mark within 3 mm of the lesion for* 6.09 *percentage points (pp) more insertions than the rotated one (95% CI* $[4.38, 7.78]$*) on a fresh LUNA16 holdout (170 patients, 12,240 insertions). The gain was* 2.76 pp *(*$[1.77, 3.76]$*) on 72 scans with manually annotated native vessels. In an earlier low-texture study, alignment also raised the share of accepted lesion crops by* 9.37 pp *(*$[4.56, 12.29]$*), but on the broader holdout this crop-level gain was smaller and its interval included zero. Plain subtraction of the forecast localized as many lesions as fitting the vessel amplitude jointly with the lesion. Correct placement of a vessel forecast therefore mainly changes* where *a detector looks, at an unchanged false-alarm budget.*

1. Introduction

Candidate scorers for chest CT look for compact, roughly spherical bright responses. Pulmonary vessels produce many of these: a vessel cut across, a bend, or a junction all give a compact peak (Fig. 1a). A scorer then faces two separate decisions. It must decide whether a crop contains a lesion, and it must say *where* in the crop to look. A vessel can spoil both. It can make a lesion-free crop look positive, and it can win the search inside a crop that does contain a lesion, so that the mark sent to a reviewer sits on the vessel.

The vessels that pass through a candidate region are usually visible as they enter and leave it. A natural idea is therefore to predict, or *forecast*, each vessel's continuation through the region from the tissue outside it, and to explain away the predicted vessel before searching for a lesion (Fig. 1c). Showing that this works requires more care than it seems. A forecast adds extra input and extra capacity, and it carries information about the type of background in the crop. Any of these could raise a score even if the forecast were in the wrong place.

We therefore test one specific claim: *the forecast helps because it is aligned with the anatomy*. Two design choices make this testable. First, each control crop and its lesion-bearing copy are identical outside a 14-mm ball, and the lesion lies inside it. The forecast, which only sees the outside, is the same for both (Fig. 1b). Second, a sham forecast is made by rotating the same field by a quarter-turn about the slice axis. The sham keeps every value, scale and smoothness of the forecast but no longer matches the vessels (Fig. 1c, right). A third control gives the classifier only summary statistics of the exterior.

We also ask how much freedom to give the vessel estimate. The same local readout is run with four treatments of the vessel amplitude, from none to a joint fit with the lesion (Sec. 4). Our contributions are:

- a paired test that separates the effect of a forecast's *placement* from the information it carries, at a fixed candidate false-positive budget;
- a shared-readout comparison showing that subtracting the forecast at a fixed amplitude is as good as fitting the amplitude jointly with the lesion;
- a localization analysis on a fresh LUNA16 holdout and on an independent dataset with annotated native vessels, showing that alignment mainly improves where the detector looks.

2. Related work

Vessels as explanations for compact responses. Multi-scale Hessian filters describe local curvature and separate tubular from compact structures [5, 8, 10]. A junction can still produce curvature along several axes, and the strongest response in a search region can lie away from the lesion. Agam *et al*. used reconstructed vessel trees to resolve such local ambiguities in CT [1]. Charbonnier *et al*. used vessels outside a subsolid nodule contour to guide vessel segmen-

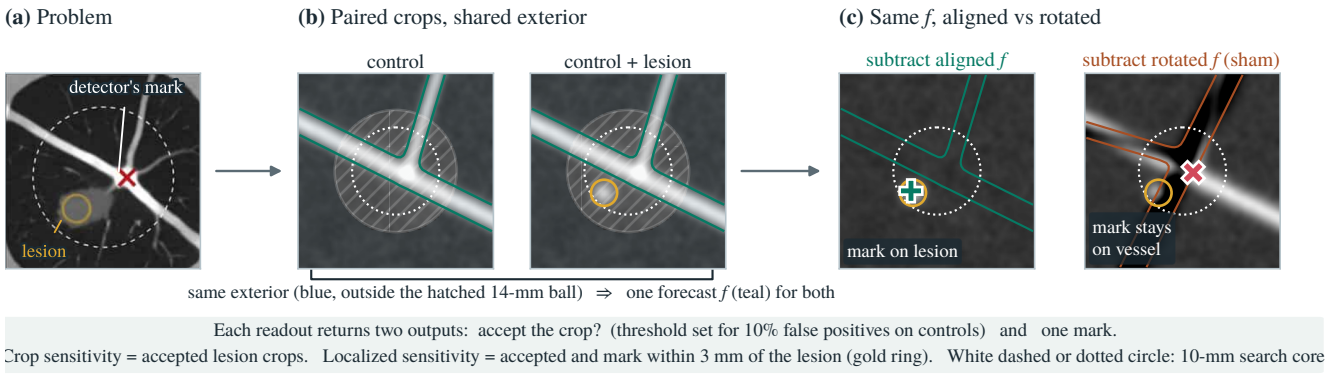


Figure 1. **Problem and idea.** (a) A compact-blob search inside the 10-mm core marks the bright vessel junction (red cross) instead of the nearby lesion (gold ring). (b) Each control crop and its lesion-bearing copy are identical outside 14 mm. The forecaster sees only this exterior and continues the vessels across the hidden ball, so both crops get the same forecast f. (c) Subtracting the aligned f removes the vessel and the search returns the lesion (green cross). The rotated sham has the same values in the wrong place: the vessel stays and keeps the mark. Panel (a) was created in BioRender; (b)–(c) use a generated slice with the same layout, and their marks are computed by a simplified version of the readout of Sec. 4.

tation inside it [3]. Learned candidate systems use local anatomy or multiple views [12, 14], and Gu *et al*. added a vessel-suppression branch to a nodule detector [6]. These works motivate using context. We instead hold the exterior signal fixed and change only its spatial alignment with the candidate.

Task-based assessment of image operations. A gain in pixel fidelity need not improve detection. Li *et al*. compared denoisers across detection observers, and Yu *et al*. found that image-quality measures and detection efficiency can move in opposite directions [7, 15]. Here the operation is vessel subtraction or conditioning. We judge it by sensitivity at a fixed false-positive budget and by whether the returned location is on the inserted target.

Detection with structured nuisance. Testing for a non-negative signal in the presence of a structured interference subspace is a classical problem [11]. We use a weighted affine background and a single exterior-derived nuisance column. Our comparison concerns where that column comes from, how its amplitude is treated, and how this changes a searched candidate score.

3. A controlled test of alignment

Crops and decisions. Each input is a CT crop y of 64^3 voxels at $1.5 \times 0.7 \times 0.7\,\mathrm{mm}$ (slice, row, column) around a proposal centre c. We use three regions: the *core* within $10\,\mathrm{mm}$ of c, where lesions are searched; the *hidden ball* within $14\,\mathrm{mm}$; and the *exterior* E beyond $14\,\mathrm{mm}$. A scorer makes two decisions. The *crop decision* $A(y) \in \{0, 1\}$ accepts the crop for review. The *location* $L(y)$ is the single point it marks.

Paired insertions. Let b be a CT background, v an optional added vessel field, and s an inserted lesion with $s = 0$ on E. The control and lesion crops are

$$y_0 = b + v, \qquad y_1 = b + v + s. \tag{1}$$

For native-vessel backgrounds $v = 0$, since the vessels are already in b. Because y_0 and y_1 agree on E, the forecast $f = F(y|_E)$ is identical for the control and all of its insertions. All readouts score y_0 and y_1 at the same centre c. The scorer itself may use the whole crop; only the forecast is restricted to E. A score can therefore rise because a lesion is present, but never because the lesion changed the forecast.

Placement sham and context control. Let Q be a $90°$ rotation about the slice axis through c. The row and column spacings are equal, so Q permutes voxels without interpolation. The sham forecast Qf keeps the radial profile, the set of values, the smoothness and the shape of f, but generally moves its vessels to where the anatomy has none. We compare the aligned arm with (i) a rotated arm whose classifier is refitted on Qf, which lets the sham make the best use of a misplaced field, and (ii) the aligned classifier applied to Qf, which removes any effect of refitting. A third control receives four exterior summary statistics but no spatial field. A gain over all three controls is attributed to placement.

4. Readouts on a shared window bank

Forecasting from two visible legs. Peaks on a sphere of radius 15.5 mm propose pairs of entry and exit points. A pair is kept if the signal continues against its transverse flanks at radii 17.8 and 20.4 mm on both sides. For each kept chord we fit a Gaussian cross-section (width and gain) on the two exterior legs, interpolate width and amplitude along the chord, and taper the field smoothly beyond 22 mm. The result f is an additive estimate in CT units. A crop without an accepted chord has $f = 0$ and is scored by the same procedure.

Concretely, we smooth the masked crop with a 0.85 mm Gaussian normalized by the smoothed visibility mask and sample it at 2,400 nearly uniform anchors on the 15.5 mm sphere. Up to 14 prominent peaks at least 4.1 mm apart are paired when they are at least 27.3 mm apart, and up to three disjoint pairs with sufficient bilateral contrast are kept, preferring agreement and proximity to the centre. Each leg is fitted on exterior samples within 5.5 mm of the chord and 11–21 mm from the centre, choosing the width from $\{1.15, \dots, 2.9\}$ mm by the largest reduction of the residual sum of squares; a track needs a valid fit on both legs and an amplitude of at least 77 HU. The taper runs from 22 to 26 mm with a cubic smoothstep. Three checks guard the shared-exterior property: the rotated field has the same value multiset on the central 13-mm support of every control; re-running the forecaster after replacing the hidden region gives an identical field; and the visible exterior of every rendered pair agrees bitwise. A straight chord is an approximation: curved paths, branches and other bright structures can disagree with it, which the native-vessel transfer tests directly.

Local windows. For each search centre x in the core, let W_x be Gaussian weights ($\sigma = 3.4$ mm) on a 6 mm ball, with inner product $\langle a, b\rangle_W = \sum_p W_x(p)\, a(p)\, b(p)$. Let B span a constant and the three coordinates in the window. We project the observation, the forecast and a compact template onto the W_x-orthogonal complement of B:

$$z = P_x y, \qquad u = P_x f, \qquad t = P_x T_{x,r}, \tag{2}$$

with templates $T_{x,r}(p) = [1 - \|p - x\|^2/r^2]_+^{0.8}$ and $r \in \{1.85, 2.7, 3.7\}$ mm. The projection removes the background level and linear drift. The local model is

$$z \approx g\, u + h\, t, \qquad h \geq 0, \tag{3}$$

where g is the vessel gain and h the lesion amplitude.

The no-lesion fit. Under the null ($h = 0$) we fit the gain with a penalty that keeps it near the exterior value $g = 1$:

$$J_0 = \min_g \|z - gu\|_W^2 + \lambda(g-1)^2, \qquad \lambda = \rho\|u\|_W^2. \tag{4}$$

With $G = \|u\|_W^2 + \lambda$, the minimizer is $g_0 = (\langle u, z\rangle_W + \lambda)/G$.

Four treatments of the vessel gain. Adding the lesion term reduces the loss by $[a]_+^2/d$, where a is the residual correlation with the template and d an effective template energy. The four readouts differ only in these two quantities:

- *CT only:* $u = 0$, so $a = \langle t, z\rangle_W$ and $d = \|t\|_W^2$.
- *Fixed:* $g = 1$ from the exterior fit, so $a = \langle t, z - u\rangle_W$, $d = \|t\|_W^2$, and $J_0 = \|z - u\|_W^2$.
- *Plug-in:* $g = g_0$ is held fixed, so $a = \langle t, z\rangle_W - g_0\langle t, u\rangle_W$ and $d = \|t\|_W^2$.
- *Joint profile:* g and h are fitted together. Eliminating g gives

$$J(h) = J_0 - 2ah + h^2 d_{\text{prof}}, \quad d_{\text{prof}} = \|t\|_W^2 - \frac{\langle t, u\rangle_W^2}{G}, \tag{5}$$

with the same a as plug-in.

Profile and plug-in coincide when template and forecast are orthogonal after projection. Where they overlap, $d_{\text{prof}} < d$, so profiling gives more weight to the same residual. The gain choice matters most at the lesion. If $z = z_0 + h_\star t$, the plug-in gain g_0 moves by $h_\star\langle u, t\rangle_W/G$, so the plug-in correlation grows by only $h_\star(\|t\|_W^2 - \langle u, t\rangle_W^2/G)$, whereas the fixed correlation grows by the full $h_\star\|t\|_W^2$. A gain fitted inside the core can absorb part of the lesion. A fixed gain cannot, but it keeps any amplitude error of the exterior fit.

Each readout has three coupled settings $(\rho, \tau, C_{\log}) \in \{(.025, .018, .008), (.15, .033, .08), (.75, .055, .8)\}$ for the gain penalty, the noise floor (in normalized CT units) and the ridge strength of the classifier; fixed subtraction ignores ρ. Numerical floors of 10^{-9} on G and 0.05 on d_{prof} keep the scores finite. For an empty forecast all four rules give the same local test before the classifier, so any difference between them on such crops comes only from the fitted coefficients.

Normalized score, search and classifier. At each centre and radius the score is

$$m_r(x) = \frac{[a]_+/\sqrt{d}}{\sigma_0}, \qquad \sigma_0 = \sqrt{J_0/\textstyle\sum_p W_x(p) + \tau^2}, \tag{6}$$

the square root of the loss reduction, scaled by the readout's own null residual with floor τ. Algorithm 1 summarizes the procedure. The three maxima $\max_x m_r(x)$ enter a ridge-logistic classifier, and the overall maximum over x and r gives the location $L(y)$. All classifiers share 12 local intensity and Hessian summaries, three template maxima from a larger window and three CT-only maxima. The shell control adds four exterior summaries, and each vessel-conditioned readout adds its three conditioned maxima. Fixed, plug-in

Algorithm 1 Forecast-conditioned candidate scoring

Require: crop y, centre c, gain rule $\in$ {CT only, fixed, plug-in, profile}
$f \leftarrow F(y|_E)$ ▷ forecast from voxels beyond 14 mm
for centres x within 10 mm of c, radii $r \in \{1.85, 2.7, 3.7\}$ mm **do**
  $z, u, t \leftarrow P_x y,\ P_x f,\ P_x T_{x,r}$ ▷ affine residuals
  $J_0, g_0 \leftarrow$ no-lesion fit, Eq. (4) (fixed: $g = 1$)
  $a, d \leftarrow$ correlation and energy for the gain rule
  $m_r(x) \leftarrow [a]_+ / \left(\sqrt{d}\,\sigma_0\right)$ ▷ Eq. (6)
end for
$\mathbf{q} \leftarrow$ shared CT features $\oplus\, (\max_x m_r(x))_r$
$A(y) \leftarrow [\mathrm{logit}(\mathbf{q}) > \kappa]$ ▷ κ at 10% FPR
$L(y) \leftarrow \arg\max_{x,r} m_r(x)$
return $A(y)$, $L(y)$

Role	Blocks	Controls	Insertions
Focused fit	46	138	2,484
Focused development	39	117	2,106
Focused calibration	92	276	–
Focused evaluated	183	549	9,882
Broader fit	80	240	4,320
Broader development	85	255	4,590
Broader calibration	175	700	–
Broader holdout	170	680	12,240
HiPaS development	24	144	2,592
HiPaS transfer	72	432	7,776

Table 1. **Cohorts.** Blocks are LUNA16 patients or HiPaS scans; each control is one negative decision and carries 18 paired insertions.

and profile readouts thus have identical feature counts and regularization budgets (Supp. S2).

Comparators. A Hessian comparator subtracts the same field at strength 0.35, 0.7 or 1.0 before a standard compact-curvature statistic at Gaussian scales of 1.3, 2.5 and 4.0 mm; the focused development data selected full subtraction. It shows how much a different readout adds when the nuisance estimate is fixed. A hybrid CNN with a small 3D feature extractor and a shared CT feature path takes as second input either smoothed CT, the aligned forecast, the rotated forecast or the residual $y - f$, with the same architecture, data, seeds and budget (Supp. S6).

5. Data and evaluation protocol

Three settings. The *focused* LUNA16 study [2, 13] has 549 neighbourhoods from 183 held-out patients, each with a single, crossing or skew added vessel and 18 insertions. Its core intensity SD is capped at 42 HU to isolate cases where vessels compete strongly with the lesion. Its original thresholds and analysis were locked before testing; our new shared-readout results on these cases are retrospective development evidence. The *broader* LUNA16 holdout uses a new, disjoint roster of 170 patients with four parenchymal neighbourhoods each (680 controls, 12,240 insertions), no texture cap (median core SD 117 HU), tubes that bend and taper, and new lesion shapes (bilobed and part-solid-like) at contrasts of 84, 196 and 364 HU and offsets of 0, 4 and 7 mm from the proposal centre. These shapes differ from the search templates. The *HiPaS* transfer [4] uses 72 noncontrast scans from a different source with manual artery and vein labels, six neighbourhoods per scan between nearby vessels (432 controls, 7,776 insertions) and no added tubes. The annotations only define the sampling population and the analysis strata; the scorer sees the image alone. Table 1 lists every role.

Sampling and lesion bank. For the broader study, every patient used in any focused role or in the broader fitting and development folds is excluded, and a fixed permutation assigns the rest to disjoint calibration and test rosters (175 of 177 and 170 of 174 patients are eligible). Candidate centres are at least 38 mm apart, more than 27 mm from any reference or irrelevant LUNA16 finding, and at most 12% of the 26-mm support lies above -552 HU. The broader holdout's core SD spans 59–255 HU between its 10th and 90th percentiles (HiPaS: 123–407 HU, median 253 HU), against the focused cap of 42 HU. The new lesions are built from an anisotropic component $L(p) = \exp(-q^2)(1 - q^2)_+^{0.6}$ with ellipsoidal radius q: a bilobed shape adds two translated components (weights .72 and .52) and a part-solid-like shape adds a broad halo and a smaller core (.40 and .60), with a random rotation and axis stretch shared across a crop's bank. The focused study used $(1 - d^2/r^2)_+^{0.72}$ with radii 1.85 and 3.35 mm at 70, 182 and 336 HU. HiPaS centres are midpoints between annotated artery and vein surfaces 2.5–8 mm apart, with annotated foreground in the core.

Fitting and pre-registration. Classifiers for the broader study are fitted on 80 patients and tuned on 85 others, with three coupled settings of ridge strength, floor τ and gain penalty ρ. Before rendering the test images we froze the cohort, the insertion bank, the analysis, and three arms chosen on development: the lead (aligned profile), the strongest alternative residual (fixed) and the refitted rotation. The primary criterion was a crop-sensitivity gain of at least 5 pp at 10% FPR, with a positive lower confidence limit against both the rotation and the fixed residual. Localized sensitivity was a prespecified secondary endpoint. A development block bootstrap estimated that about 123 patients give 80% power to detect a true 5-pp crop gain over the rotation, and 74 over fixed subtraction; the 170 test patients exceed both.

Readout	Focused sens.	Broader LUNA16 holdout sens.	loc. sens.	frozen
Fixed, aligned	50.75	24.05	15.38	24.74 / 10.59
Fixed, rotated	36.03	20.81	5.72	21.18 / 10.15
Plug-in, aligned	39.45	22.54	11.54	23.86 / 10.44
Plug-in, rotated	37.39	21.26	6.92	19.81 / 9.12
Profile, aligned	**46.50**	**23.32**	**13.45**	**23.79 / 10.59**
Profile, rotated	37.13	21.23	7.35	19.29 / 8.82
Prof. rot., aligned clf.	–	19.26	9.19	19.20 / 10.00
CT + shell	36.60	21.50	13.49	21.66 / 10.29
CT only	32.88	21.39	13.37	23.78 / 11.62
Hessian subtr.	47.34	14.31	–	14.31 / 10.29
Lead minus … (pp, 95% CI)				
− refitted rotation	–	+2.1 [−0.3, +4.4]	+6.1 [+4.4, +7.8]	–
− fixed, aligned	–	−0.7 [−3.0, +0.5]	−1.9 [−3.3, −1.0]	–

Table 2. **Placement and amplitude comparisons.** Sensitivity (%) at the matched 10% FPR. "Focused" reuses the focused study for development; the broader holdout uses separate fits. "loc. sens." requires acceptance and a peak within 3 mm (Eq. 7). "frozen" gives sensitivity / achieved FPR at the independent LUNA16 cutoffs. Bold: the lead arm chosen before testing.

Operating points and endpoints. We report two operating points. The *matched* point sets each arm to a 10% candidate FPR on the evaluation controls, using the order statistic of their scores; it measures ranking at a fixed budget. The *frozen* point applies cutoffs set on 700 independent negative crops from 175 LUNA16 calibration patients, accepting scores strictly above the 10% order statistic; its achieved FPR can differ between arms. *Localized sensitivity* is

$$\Pr\big(A(y) = 1,\ \|L(y) - x_\star\| \le 3\,\mathrm{mm} \mid Y = 1\big), \quad (7)$$

where $x_\star$ is the insertion centre. Controls still count against the FPR budget whether or not their mark is meaningful.

Uncertainty. We resample 3,000 sets of patient blocks (scans for HiPaS), stratified by release fold, keeping each block's full insertion grid and controls, so all comparisons are paired. For matched intervals the cutoff is recomputed in every draw. For frozen cutoffs we report intervals conditional on the calibration set and, separately, intervals that also resample the calibration patients. Differences are in absolute percentage points.

6. Results

Alignment, in the focused study. In the original locked study, the aligned profile beat its rotation by $+9.66$ pp [+8.10,+11.19] at frozen cutoffs (achieved FPR 10.38% vs. 10.02%) and by $+9.37$ pp [+4.56,+12.29] at matched FPR (Fig. 2b). With the shared readout on the same cases, fixed subtraction gained $+14.72$ pp [+11.11,+17.05] over its rotation and $+14.15$ pp [+10.73,+16.44] over shell summaries, gaining 2,001 and losing 546 positive decisions relative to the rotation (Table 2).

Alignment, in the broader holdout. The broader holdout is harder: every arm's crop sensitivity is near 21–24%. Here the aligned profile raised crop sensitivity by only $+2.08$ pp [-0.27,+4.40] over the refitted rotation and by $+4.05$ pp [+1.37,+6.23] over the rotation with the aligned classifier. At frozen cutoffs the difference was $+4.50$ pp, but at achieved FPRs of 10.59% vs. 8.82%; with calibration uncertainty included the interval was [-0.59,+8.31] pp and the excess-FPR upper limit 5.00 pp. **The prespecified crop criterion was therefore not met.** Localization tells a different story (Fig. 3). The rotated profile marked the lesion in 7.35% of insertions; the aligned profile in 13.45%, a gain of $+6.09$ pp [+4.38,+7.78].

How alignment changes the mark. Comparing the two arms crop by crop shows where this gain comes from. Among crops accepted by *both* arms, alignment moved 386 marks from off target to within 3 mm and 124 the other way. A further 647 crops were rejected by the rotation and accepted and localized by the aligned arm, against 163 in the reverse direction (Supp. Tab. S2). The net change of 746 localized insertions is much larger than the change in accepted crops because many gains happen on crops that were already accepted. The mechanism is the competition shown in Fig. 1: if a vessel response $m(x_v)$ leads a lesion response $m(x_\ell)$ and conditioning lowers them by d_v and d_ℓ, the lesion takes the lead when $d_v - d_\ell > m(x_v) - m(x_\ell)$, whether or not the crop decision changes.

How much amplitude flexibility is needed? More flexibility did not help. In the focused study the original profile was -0.84 pp [-6.15,+2.43] below the Hessian comparator, while the matched fixed residual was $+3.41$ pp [-0.23,+6.05] above it. In the broader holdout the profile was -0.74 pp [-3.04,+0.54] below fixed subtraction in crop sensitivity, and fixed subtraction localized $+1.93$ pp [+1.05,+3.27] more insertions (15.38% of all insertions). With exact fields on 18 development backgrounds, where the forecast equals the added vessel, fixed subtraction and profiling both ranked 67.6% of insertion/control pairs correctly, and plug-in 62.0%. Fitting the gain inside the core lets part of the lesion be absorbed, as predicted in Sec. 4.

A competent learned comparison. The observed-CT CNN reached 67.54% development AUROC and 36.28% sensitivity at the 10% cap, close to the local ridge classifier (68.08% and 36.47%). On the focused cases its aligned-minus-rotated difference was -0.54 pp [-2.80,+0.62], and

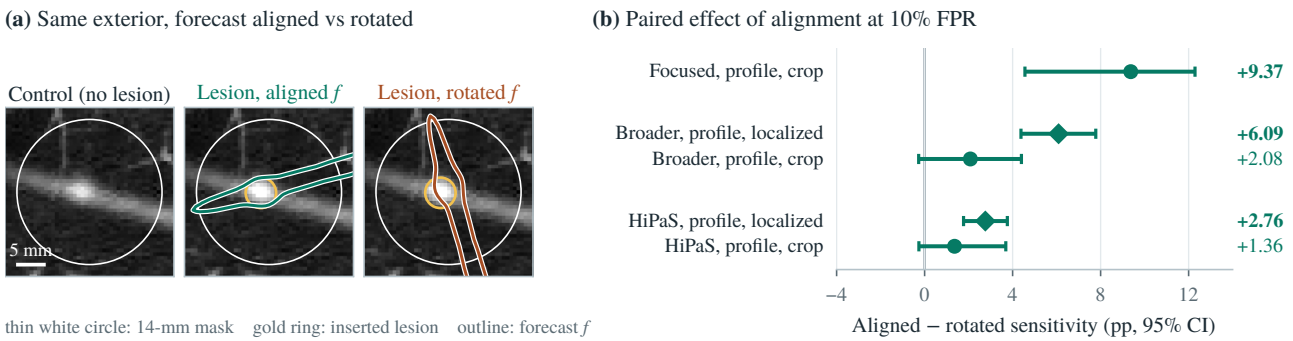


Figure 2. **The same forecast helps only when it is placed correctly.** (a) A development example: the control (left) and its lesion-bearing copy with the aligned (middle) and rotated (right) forecast outline. The thin white circle is the 14-mm mask and the gold ring the inserted lesion; note that the control already shows a compact vessel response near the insertion site. All panels share one display window; the returned marks of this example are not shown. (b) Aligned minus rotated sensitivity of the joint-profile readout at the matched 10% FPR. Circles: crop sensitivity; diamonds: localized sensitivity. Bold values: interval excludes zero. In all figures, intervals are 95% paired block-bootstrap CIs (3,000 draws of patients, or scans for HiPaS).

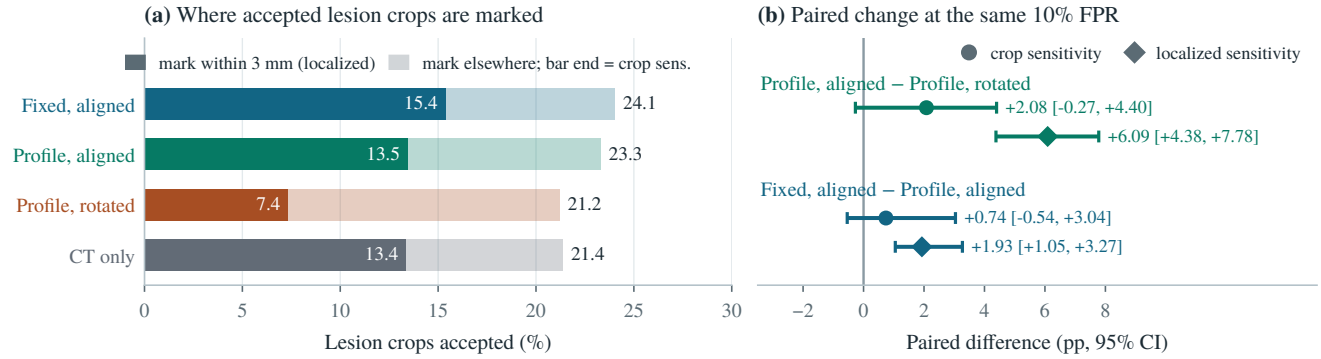


Figure 3. **Acceptance and localization are separate decisions** (broader holdout, matched 10% FPR). (a) Accepted lesion crops, split into those whose mark lies within 3 mm of the insertion (dark; localized sensitivity) and elsewhere (light). The bar end is the crop sensitivity. All readouts use the same search and the same lesion crops. (b) Paired differences (circles: crop, diamonds: localized sensitivity) with their 95% CIs printed.

the residual channel added +2.16 pp [-1.19,+4.44]. Table 3 lists all four inputs. Before the comparison we checked that the CNN can learn: on four fit patients it reaches 99.48% training AUROC with strong off-centre positives and 86.42% with all insertion variants, and its development AUROC rises with the number of fit patients (Fig. 4). A classifier that pools features can therefore use the same field without exploiting its placement. The placement gain belongs to a readout that lets the field remove a competing peak.

Second input	Epoch	Dev. AUROC	AUROC	Sens.
Smoothed CT	30	67.54	68.78	34.28
Aligned forecast	20	67.54	68.81	35.14
Rotated forecast	30	67.81	69.13	35.68
Residual $y - f$	30	68.16	69.46	36.44

Table 3. **Hybrid CNN with four second inputs** (%). Same architecture, data, seeds and budget; the epoch is selected on development. AUROC and sensitivity (10% FPR cap) are on the focused cases; the CT-only ridge reaches 69.17% and 32.88%.

7. Where alignment helps

Competition inside the search region. Fig. 5 splits the paired crop-level gain by insertion condition, always at each study's global cutoff. Gains concentrate where a vessel is likely to win the search: displaced, high-contrast insertions. In the broader holdout these gained +15.40 pp [+11.84,+19.08]. Centred insertions lost −5.34 pp [-9.49,-1.25], faint ones −1.47 pp [-2.60,-0.24], and insertions overlapping a vessel by at least 50% were unchanged (−0.12 pp [-3.59,+3.23]). A centred lesion often already is the strongest response, and a faint one leaves a weak residual even when the vessel is explained well. The focused study shows the same pattern with larger effects. A forecast-dominance score

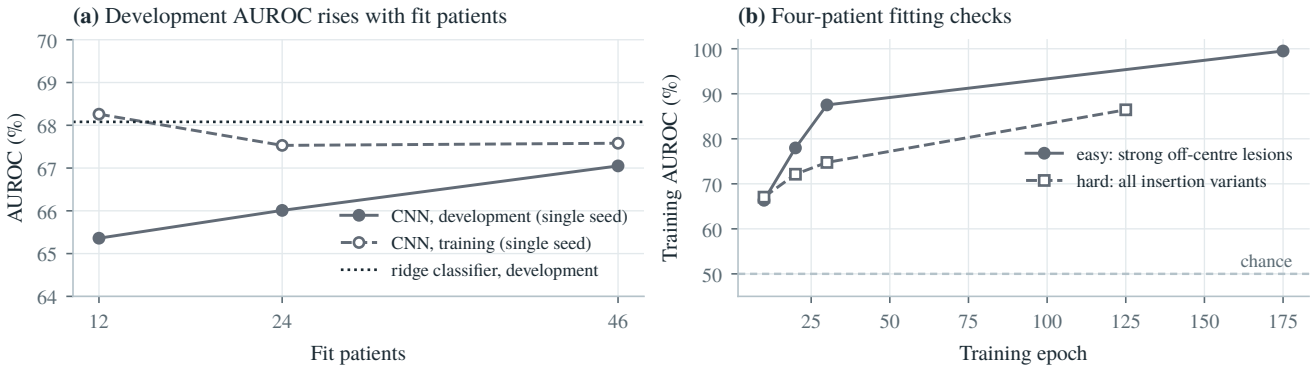


Figure 4. **The learned comparison is a working classifier.** (a) Observed-CT CNN AUROC versus the number of fit patients (single-seed snapshots; the reported 67.54% averages two seeds), with the local ridge on development for reference. (b) Training AUROC on four fit patients for an easy case (strong off-centre lesions) and a hard case (all insertion variants); the hard probe was recorded up to 125 epochs.

that can be computed before scoring (peak forecast relative to exterior texture; Supp. S5) selects 40.7% of broader crops, where the gain is +3.29 pp [+0.04,+6.03]; the high-minus-low interaction, +2.03 pp [-1.62,+5.11], is not resolved. The score is the peak forecast within 13 mm divided by 0.02 plus the 80th–20th percentile spread of the exterior residual in the 15.5–19.5 mm annulus, split at the development median. The high band holds 277 controls and 4,986 insertions; there the aligned and rotated profiles have frozen FPRs of 11.19% and 9.39% and miss 3,731 and 4,009 insertions. Outside it, the rates are 10.17% and 8.44% with 5,597 and 5,870 misses, so the score describes where alignment matters but does not on its own select a better operating point.

Cross-checks of the localization gain. The lead comparison uses the preselected profile. As a post hoc check, fixed subtraction against its own separately tuned rotation gains +9.66 pp [8.24,11.38] in localized sensitivity on the broader holdout and +2.39 pp [1.22,3.28] on HiPaS. Holding the aligned classifier fixed and only rotating the field gives +4.26 pp [2.39,5.83] and +0.51 pp [-0.15,1.09], respectively. The gain is therefore not specific to one readout, and on the broader holdout it survives even when the classifier is not refitted.

Native vessels. On the 72 HiPaS scans, 7,713 of the 7,776 insertions lie within 2 mm of an annotated artery or vein. At matched FPR, alignment improved localized sensitivity by +2.76 pp [+1.77,+3.76] and crop sensitivity by +1.36 pp [-0.26,+3.69]. The localization gain therefore persists when real anatomy, including curved vessels that a straight chord fits poorly, supplies the competing response. The aligned fixed and profile readouts localize about twice as many insertions as the rotation (Fig. 7a), and alignment produced 284 newly localized marks against 69 lost.

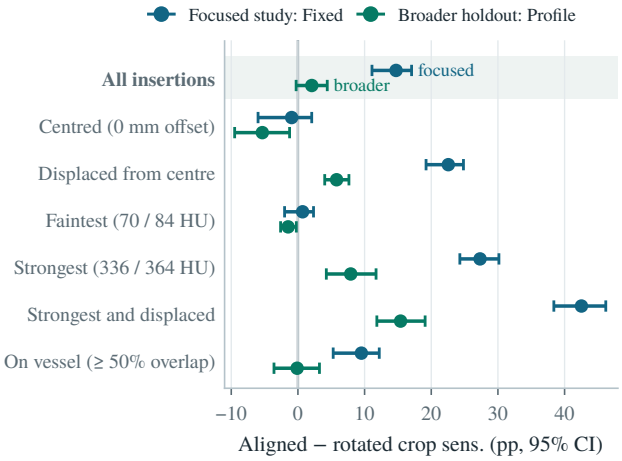


Figure 5. **Placement gains by insertion condition.** Aligned minus rotated crop sensitivity at each study's global 10% FPR cap. Blue: fixed subtraction in the focused study (development cases); green: the prespecified joint profile in the broader holdout. Contrast values are focused / broader. Rows are not mutually exclusive.

Thresholds do not transfer. Applied unchanged to HiPaS, the LUNA16 cutoffs gave the aligned profile 2.15% sensitivity at 0.69% FPR, versus 6.62% at 4.63% for its rotation (Fig. 7b). Most readouts accepted far fewer HiPaS controls than the nominal 10%, so their realized operating points moved. The ranking result and the transferred decision rule therefore answer different questions, and a deployment claim would need negatives from the target acquisition.

Ranking versus calibrated cutoffs. Figure 6 shows the full ROC curves behind these operating points. On the broader holdout the two aligned readouts lie above the rotation and the CT-only classifier from about 3% to 25% FPR, so the matched result does not hinge on the 10% cap;

	Matched 10% FPR		LUNA16 cutoffs
Readout	sens.	loc. sens.	sens. / FPR
Fixed, aligned	16.26	5.66	1.93 / 0.46
Fixed, rotated	14.30	3.27	5.00 / 2.55
Plug-in, aligned	15.64	5.31	2.31 / 0.69
Plug-in, rotated	14.63	2.84	6.53 / 3.94
Profile, aligned	**15.52**	**5.47**	**2.15 / 0.69**
Profile, rotated	14.16	2.70	6.62 / 4.63
Profile, rot., aligned clf.	14.97	4.95	0.91 / 0.23
CT + shell	14.45	4.21	4.69 / 2.55
CT only	14.47	3.38	10.83 / 7.87
Hessian subtraction	11.33	–	12.04 / 10.88

Table 4. **HiPaS transfer** (432 controls, 7,776 insertions; %). Matched columns use the within-HiPaS 10% FPR cap; the last column applies the independent LUNA16 cutoffs. Controls contain native anatomy and no inserted target.

Contrast	Intervals	Sensitivity Δ	FPR Δ
Broader LUNA16			
Lead−rot.	frozen	+4.50 [+2.62, +6.32]	+1.76 [−0.59, +4.12]
	recal.	+4.50 [−0.59, +8.31]	+1.76 [−2.35, +5.00]
Lead−fixed	frozen	−0.95 [−1.85, −0.10]	+0.00 [−1.32, +1.32]
	recal.	−0.95 [−3.80, +1.32]	+0.00 [−2.21, +1.76]
HiPaS			
Lead−rot.	frozen	−4.48 [−6.51, −2.58]	−3.94 [−6.02, −2.08]
	recal.	−4.48 [−7.09, −2.60]	−3.94 [−6.02, −1.62]
Lead−fixed	frozen	+0.22 [−0.05, +0.62]	+0.23 [+0.00, +0.69]
	recal.	+0.22 [−0.32, +0.73]	+0.23 [+0.00, +0.69]

Table 5. **Paired differences at the frozen LUNA16 cutoffs** (pp, 95% CI). "frozen": conditional on the calibration set; "recal.": calibration and evaluation blocks both resampled. The lead is the aligned profile.

on HiPaS the crop-level curves nearly coincide and the gain appears only in localization. The frozen LUNA16 cutoffs (diamonds) land close to the 10% cap on LUNA16 but far below it on HiPaS. Table 5 shows how much of the frozen-cutoff uncertainty comes from the calibration set itself: resampling the 175 calibration patients widens the broader lead-minus-rotation interval from $[+2.62, +6.32]$ to $[-0.59, +8.31]$ pp. On HiPaS the lead accepts fewer insertions *and* fewer controls than its rotation, an operating-point shift rather than a ranking loss (Table 4).

8. Discussion and limitations

Implications for candidate scoring. A forecast has several ways to look useful: its shape and amplitude correlate with the type of background in a crop. The rotation sham, the shell summaries and the shared readout isolate what its *alignment* adds. We suggest two habits for work in this area: compare against a value-preserving placement sham before attributing a gain to geometry, and include plain fixed subtraction with the same search as a baseline before adding a more flexible nuisance model. When a pipeline shows a reviewer a marked location, moving that mark onto the lesion matters even if the number of accepted crops barely changes. Conversely, a gain in crop sensitivity can overstate what the mark shows when the peak stays on the vessel.

	Profile		Rotation		Hessian	
Condition	Sens.	FPR	Sens.	FPR	Sens.	FPR
Focused LUNA16	46.85	10.38	37.19	10.02	38.39	5.46
Morphology shift	41.79	12.05	32.69	9.24	32.06	6.43
LNDb + morphology	28.04	6.49	29.27	9.14	36.09	12.09

Table 6. **Original frozen operating points of the focused study** (%), with its own 276 LUNA16 calibration negatives. These results are separate from the new broader holdout and HiPaS transfer.

Operating points. The focused study's original primary criterion also required an excess-FPR upper limit of 4 pp relative to the CT-only classifier; the observed limit was 4.92 pp. Neither shifted condition of that study, a LUNA16 morphology shift and an independent LNDb acquisition with shifted insertions [9], met its full transfer criteria (Table 6). Their within-test profile and Hessian sensitivities at 10% FPR were 37.93% and 37.44% under morphology shift and 33.15% and 32.94% on LNDb, so ranking transferred better than the frozen cutoffs did. On LNDb, adding the spatial features to a published detector's scores changed mean FROC by 0.00 pp [-0.21,+0.23] against the full local classifier and −0.13 pp [-0.33,+0.35] against the sham.

A matched-FPR effect establishes ranking separation at a given budget; a shift in the negative score distribution changes the realized budget at an imported threshold.

Choosing when to use the forecast. A rule that falls back to the shell-only classifier when the two exterior legs disagree reached 48.9% sensitivity at 65% development coverage, against 47.9% for a forecast-strength rule and 52.1% for always using the aligned fixed residual. All three cost 132–137 ms per loaded crop on one CPU thread. We therefore make no claim about selective use.

Scope. Every positive target is an inserted signal. The broader study varies texture and morphology under a known intervention, and HiPaS adds an independently annotated native background. Real nodules, scan-level proposal pipelines, partial-volume effects and per-scan false-positive counts are outside this study; an adjudicated scan-level evaluation would be needed to quantify clinical detection.

9. Conclusion

Paired CT insertions with a shared exterior show when vessel evidence from outside a candidate region changes the decision inside it. A correctly placed forecast releases a

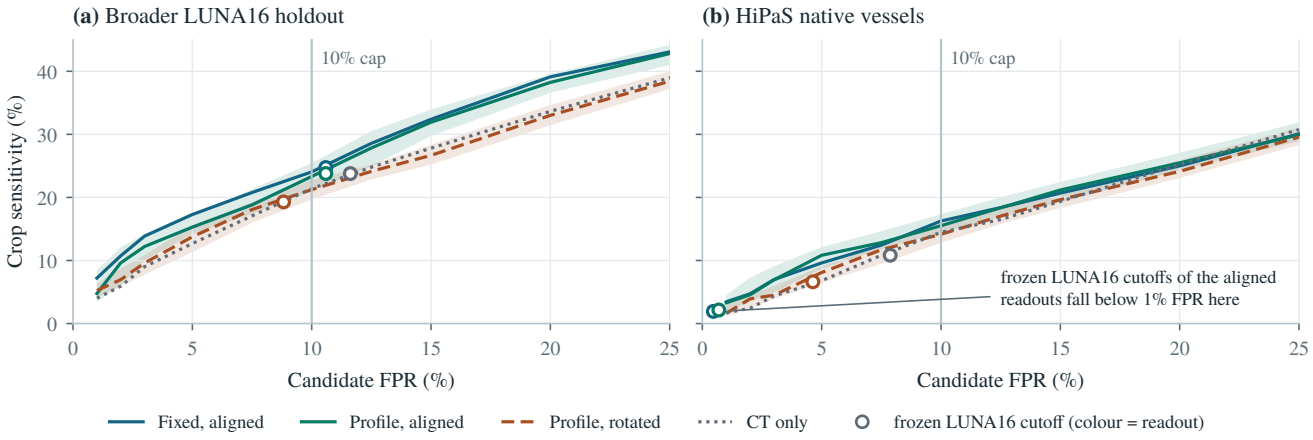


Figure 6. **ROC curves and transferred thresholds answer different questions.** Curves use within-cohort negatives; shading gives pointwise 95% block intervals for the two profile arms. Open circles: the frozen LUNA16 cutoffs, applied unchanged in both domains.

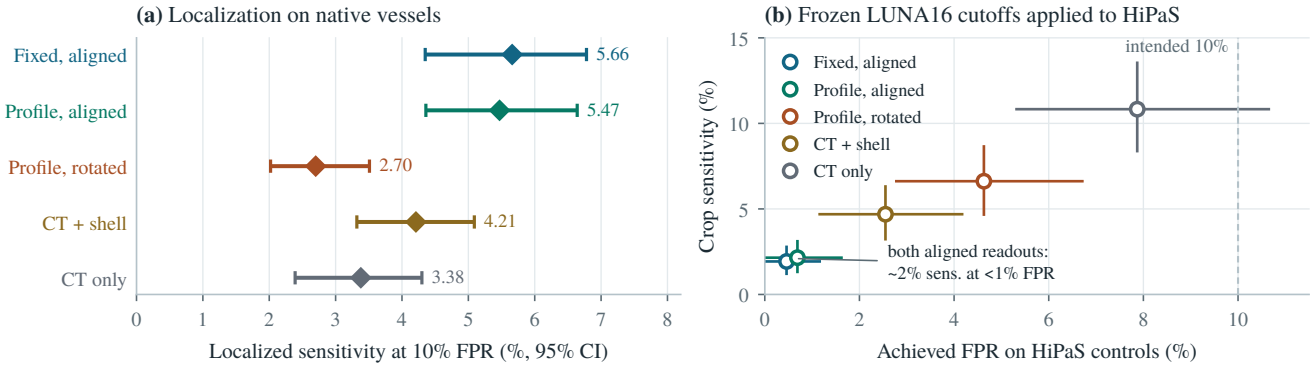


Figure 7. **Localization gains transfer to annotated native vessels; thresholds do not.** (a) Localized sensitivity on 7,776 HiPaS insertions at the within-HiPaS 10% FPR cap. (b) Crop sensitivity and achieved FPR when the frozen LUNA16 cutoffs are applied unchanged (intervals conditional on the cutoffs). Colours as in Fig. 3, with CT + shell in olive; diamonds: localized sensitivity; open circles: crop sensitivity at the frozen cutoffs, as in Fig. 6.

competing vessel peak and returns the inserted target at the same false-alarm budget, and this localization effect persists on manually annotated native vessels. Fixed subtraction with a compact-template search exploits the same alignment as well as joint profiling, which makes it a strong baseline for any more elaborate use of forecast uncertainty.

LLM Usage Statement

OpenAI Codex assisted with study design, implementation, experiment execution, numerical analysis, literature lookup, figure preparation, and manuscript drafting and editing. Claude (Anthropic) assisted with revising the manuscript text and redrawing the figures from the recorded measurements. The quantitative figures were generated programmatically from recorded measurements. The qualitative panels use public CT crops or algorithmically generated phantoms. The authors are responsible for validating the work and its interpretation.

References

[1] Gady Agam, Samuel G. Armato III, and Changhua Wu. Vessel tree reconstruction in thoracic CT scans with application to nodule detection. *IEEE Transactions on Medical Imaging*, 24(4):486–499, 2005. 1

[2] Samuel G. Armato, III, Geoffrey McLennan, Luc Bidaut, et al. The lung image database consortium (LIDC) and image database resource initiative (IDRI): A completed reference database of lung nodules on CT scans. *Medical Physics*, 38 (2):915–931, 2011. 4

[3] Jean-Paul Charbonnier, Kaman Chung, Ernst T. Scholten, Eva M. van Rikxoort, Colin Jacobs, Nicola Sverzellati, Mario Silva, Ugo Pastorino, Bram van Ginneken, and Francesco Ciompi. Automatic segmentation of the solid core and enclosed vessels in subsolid pulmonary nodules. *Scientific Reports*, 8:646, 2018. 2

[4] Yuetan Chu, Gongning Luo, Longxi Zhou, et al. Deep learning-driven pulmonary artery and vein segmentation reveals demography-associated vasculature anatomical differences. *Nature Communications*, 16:2262, 2025. 4

[5] Alejandro F. Frangi, Wiro J. Niessen, Koen L. Vincken, and Max A. Viergever. Multiscale vessel enhancement filtering. In *Medical Image Computing and Computer-Assisted Intervention*, pages 130–137, 1998. 1

[6] Xiaomeng Gu, Weiyang Xie, Qiming Fang, Jun Zhao, and Qiang Li. The effect of pulmonary vessel suppression on computerized detection of nodules in chest CT scans. *Medical Physics*, 47(10):4917–4927, 2020. 2

[7] Kaiyan Li, Weimin Zhou, Hua Li, and Mark A. Anastasio. Assessing the impact of deep neural network-based image denoising on binary signal detection tasks. *IEEE Transactions on Medical Imaging*, 40(9):2295–2305, 2021. 2

[8] Qiang Li, Shusuke Sone, and Kunio Doi. Selective enhancement filters for nodules, vessels, and airway walls in two- and three-dimensional CT scans. *Medical Physics*, 30(8): 2040–2051, 2003. 1

[9] João Pedrosa, Guilherme Aresta, Carlos Ferreira, et al. LNDb challenge on automatic lung cancer patient management. *Medical Image Analysis*, 70:102027, 2021. 8

[10] Yoshinobu Sato, Shin Nakajima, Nobuyuki Shiraga, Hideki Atsumi, Shigeyuki Yoshida, Thomas Koller, Guido Gerig, and Ron Kikinis. Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images. *Medical Image Analysis*, 2(2):143–168, 1998. 1

[11] Louis L. Scharf and Benjamin Friedlander. Matched subspace detectors. *IEEE Transactions on Signal Processing*, 42(8): 2146–2157, 1994. 2

[12] Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Geert Litjens, Paul Gerke, Colin Jacobs, Sarah J. van Riel, Mathilde Marie Winkler Wille, Matiullah Naqibullah, Clara I. Sanchez, and Bram van Ginneken. Pulmonary nodule detection in CT images: False positive reduction using multi-view convolutional networks. *IEEE Transactions on Medical Imaging*, 35 (5):1160–1169, 2016. 2

[13] Arnaud A. A. Setio, Alberto Traverso, Thomas de Bel, et al. Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The LUNA16 challenge. *Medical Image Analysis*, 42:1–13, 2017. 4

[14] Dijia Wu, Le Lu, Jinbo Bi, Yoshihisa Shinagawa, Kim L. Boyer, Arun Krishnan, and Marcos Salganicoff. Stratified learning of local anatomical context for lung nodules in CT images. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 2791–2798, 2010. 2

[15] Zitong Yu, Md Ashequr Rahman, Richard Laforest, Thomas H. Schindler, Robert J. Gropler, Richard L. Wahl, Barry A. Siegel, and Abhinav K. Jha. Need for objective task-based evaluation of deep learning-based denoising methods: A study in the context of myocardial perfusion SPECT. *Medical Physics*, 50(7):4122–4137, 2023. 2